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Research Paper

Design and additive manufacturing of 3D-architected ceramic metamaterials with programmable thermal expansion

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ABSTRACT

Metamaterials with multiple anisotropic properties have strong potential in medicine, military, and civil engineering applications. However, previous studies have been limited to the design and fabrication of polymers, metals, and alloys, which can allow the retention of the structure at low temperatures. Metamaterials that operate at high temperatures or over large temperature ranges need to be developed. In this study, 3D-architected ceramic metamaterials with programmable thermal expansion, which can be used in high-temperature or large-temperature-range environments, were designed, fabricated, and characterized. First, three multi-ceramic quadrangular-pyramid-structured metamaterials with programmable thermal expansion behaviors were proposed and designed based on ZrO_2 and Al_2O_3 ceramics. The design mechanism and programmable range of thermal expansion were analyzed. Then, 3D-architected ceramic metamaterials with negative (-10×10^{-6}/°C), zero (0×10^{-6}/°C), and positive ($+10 \times 10^{-6}$/°C) thermal expansion were designed and fabricated through stereolithography additive manufacturing. The manufactured 3D-architected multi-ceramic metamaterials with programmable thermal expansion behaviors were analyzed using a digital image correlation (DIC) method. The results showed that the experimentally measured values were in good agreement with the theoretical values. The obtained findings pave the way for the application of 3D-architected multi-ceramic metamaterials.

1. Introduction

Metamaterials with multiple anisotropic properties, including negative Poisson's ratios [1,2], negative thermal expansion [3,4], negative modulus [5], high strength and stiffness [6], and double-negative acoustic properties [7,8], have strong potential in medicine, military, and civil engineering applications [9–11]. At present, although metamaterials have been successfully proposed by structural design and can be fabricated by conventional machining [12–14] and even some additive manufacturing techniques [3, 15–17], previous works mainly focused on polymer metamaterials and metal or alloy metamaterials. For polymer metamaterials, Choi [18] fabricated 3D metamaterials from photosensitive resins using a stereolithography (SL) additive manufacturing system. Metamaterials with giant, tailorable negative Poisson's ratios were also manufactured by a digital light processing (DLP) additive manufacturing technique in Chen's work [19]. A multi-polymer metamaterial with a tunable negative coefficient of thermal expansion (CTE) over 170 °C was proposed by Wang [3]. For metals or alloy metamaterials, Zhang et al. [20] printed Mg–NiTi interpenetrating phase composite metamaterials with high strength, damping capacity, and energy absorption efficiency using the selective laser melting (SLM) additive manufacturing technique. Carroll et al. [21] fabricated multi-metallic functionally graded materials (FGMs) using laser-based directed energy deposition (DED) additive manufacturing from normal 304 L stainless steel and Inconel 625.

Metamaterials with zero or negative thermal expansion behaviors have attracted considerable attention recently [3,13,14,16,22], and they have great application potential for structures or devices working at high temperatures or over a large temperature range in aerospace and high-temperature instrument applications. For instance, the satellite antennas and their supporting structures used in earth-orbiting satellites typically cycle between direct sunlight to shadow [23]. The major

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problem for this application is that these satellites suffer a large temperature variation, affecting the accuracy of the satellite antennas and their supporting structures; therefore, metamaterials with tailorable, particularly zero, CTEs are urgently needed. To date, many types of metamaterials with tailorable thermal expansion behavior have been designed and prepared. Raminhos et al. fabricated 2D lightweight composite meshes with a combination of a negative Poisson's ratio and negative thermal expansion from Nylon–PVA and PP–CPE+ using additive manufacturing technology [16]. 3D polymer metamaterial micro-lattices with negative thermal expansion were fabricated using gray-tone laser lithography in Qu's work [22]. Steeves et al. [24] fabricated metamaterial lattices using aluminum and titanium alloy constituents, and the lattices exhibited low CTEs and high stiffness. Toropova et al. [25] designed metamaterial cell lattices with different CTEs without generating thermal stresses during temperature fluctuations using metals, including aluminum, titanium, and lead. However, these metamaterials are typically designed based on polymers or metals, which cannot work at very high temperatures, particularly for aerospace applications. Therefore, it is necessary to develop metamaterials with controllable thermal expansion behavior that can work at elevated temperatures or over large temperature variations.

Fortunately, ceramic metamaterials can meet the requirements of elevated-temperature and large-temperature-variation environments. Although materials and structures providing tunable thermal expansion (including positive, negative, or zero) have been widely studied in recent decades [26–30], little has been reported on ceramic metamaterials. In our previous study [31], 2D-architected multi-ceramic metamaterials with programmable thermal expansion, which consisted of planar lattices and triangular structures, were designed and additively manufactured from typical Al_2O_3 and ZrO_2 ceramics with positive CTE values. The CTE experimental test results were in good agreement with the theoretical design values. Until now, however, few studies have reported the design and fabrication of 3D-architected ceramic metamaterials with programmable thermal expansion.

The aim of this study was to design and produce a 3D-architected ceramic metamaterial with programmable thermal expansion, including negative thermal expansion (NTE), zero thermal expansion (ZTE), and positive thermal expansion (PTE). The theoretical design, additive manufacturing, and thermal expansion behavior of the 3D-architected ceramic metamaterial were investigated in detail. The design and additive manufacturing scheme of the 3D-architected ceramic metamaterial with programmable thermal expansion is illustrated in Fig. 1. First, 3D quadrangular pyramid multi-ceramic structures with programmable thermal expansion were designed by changing the constituents and angle. Typical ceramics with positive CTE values, Al_2O_3 and ZrO_2, were chosen as the constituent materials. Both the theoretical basis of the design and the tailorable range were analyzed. To verify the reliability of the design methods, 3D-architected ceramic metamaterials with negative, zero, and positive thermal expansion were designed and fabricated using stereolithography (SL) additive manufacturing technology. Functionally graded ceramic (FGC) layers were used to reduce the thermal mismatch stress between the two constituent ceramics. The as-prepared 3D-architected ceramic metamaterial green bodies were then debinded and sintered at high temperatures. Finally, 3D-architected ceramic metamaterials with programmable thermal expansion were obtained, which were then analyzed using a digital image correlation (DIC) method.

2. Design

2.1. Theoretical basis

First, the theoretical basis for designing 3D-architected ceramic metamaterials with programmable expansion is presented. In this study, multi-ceramic quadrangular pyramid (QP)-structured metamaterials were designed. As shown in Fig. 2, the bottom member and hypotenuse

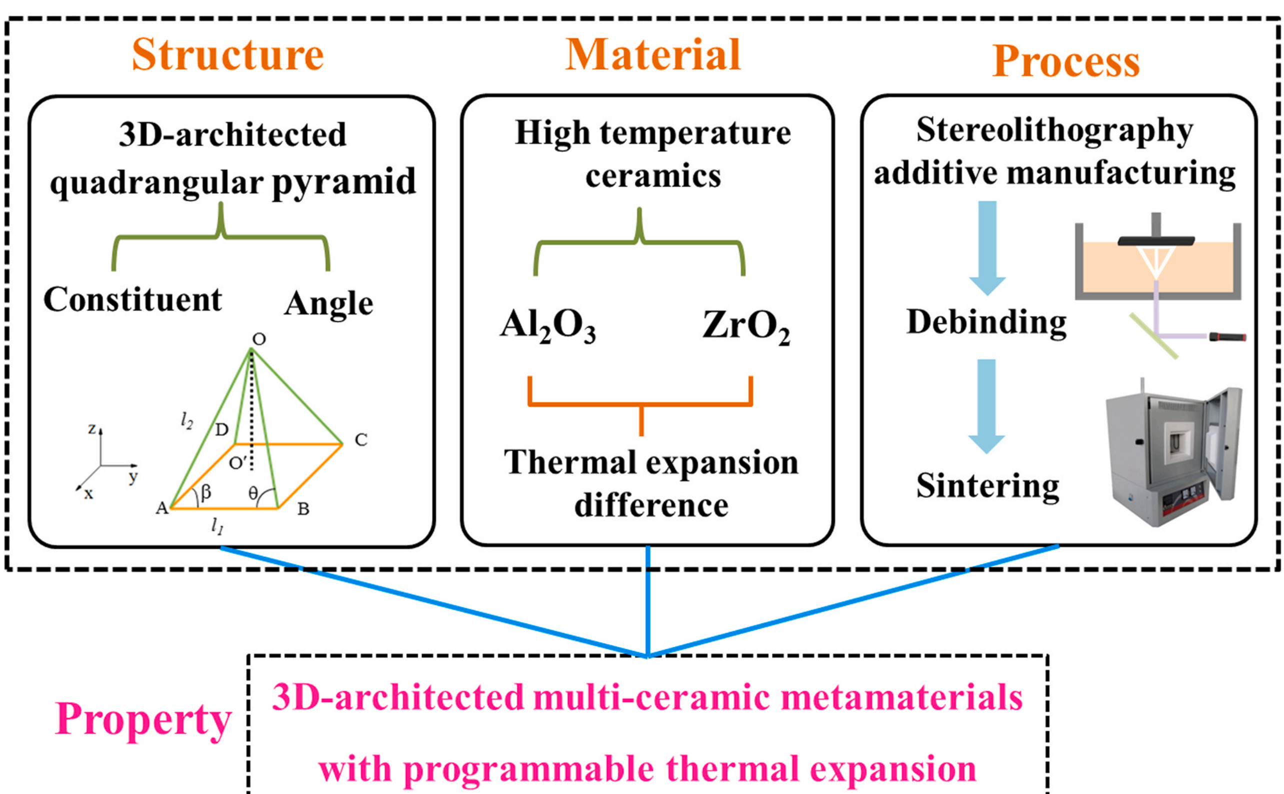

Fig. 1. Schematic of the design and stereolithography additive manufacturing of 3D-architected multi-ceramic metamaterials with programmable thermal expansion.

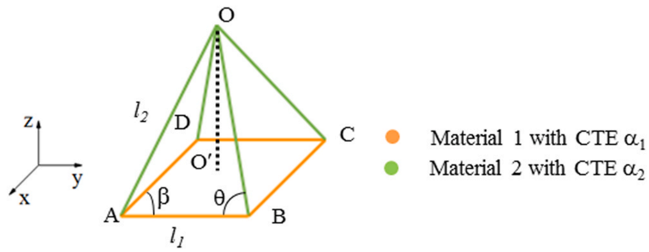

Fig. 2. Schematic of a multi-ceramic quadrangular pyramid (QP)-structured metamaterial. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

member of the multi-ceramic QP structure are l_1 and l_2, respectively. θ is the angle between bottom member l_1 and hypotenuse member l_2, satisfying $\cos\theta = l_1/2l_2$. The QP-structured metamaterial consists of two materials with different CTE values. For instance, the bottom members and the four hypotenuse members consist of material 1 with a CTE value of α_1 and material 2 with a CTE value of α_2, respectively. The bottom members expand in the horizontal direction with increasing temperature, the height of the QP structure is then forced to decrease. While, the height of the QP-structured metamaterial increases because the hypotenuse members expand in the axial direction owing to the increase in temperature. Therefore, the height of the QP-structured metamaterial is determined by the expansions of bottom members and hypotenuse members, it will decrease, have no change, or increase under different conditions. If the expansion of the bottom members is smaller than that of the hypotenuse members, the height of the QP-structured metamaterial increases and the QP-structured metamaterial exhibits PTE behavior. On the contrary, if the expansion of the bottom members is larger than that of the hypotenuse members, the height of the QP-structured metamaterial decreases and the QP-structured metamaterial exhibits NTE behavior. Specially, when the expansion of the bottom members is equal that of the hypotenuse members, the height of the QP-structured metamaterial has no change and the QP-structured metamaterial exhibits ZTE behavior. In other words, the thermal expansion behavior depends on the relative rotation between the bottom members and the hypotenuse members.

Next, three types of typical multi-ceramic QP-structured metamaterials (symmetrical) with different thermal expansion behaviors were designed, and their thermal expansion behaviors were analyzed. The as-designed multi-material QP-structured metamaterials were named QP-1, QP-2, and QP-3 for simplicity.

2.1.1. QP-1-structured metamaterial

Fig. 3 illustrates the thermal expansion of the multi-ceramic QP-1-structured metamaterial. The CTEs of the two opposite bottom sides of the QP-structured metamaterial are completely equal. Thus, with a temperature variation ΔT uniformly distributed on the QP-structured metamaterial, the bottom members expand in the horizontal direction, and rotation does not occur; the bottom square is transformed into a rectangle, and the four corners remain at 90°. Nevertheless, triangle OAB consists of two ceramics, and the CTE of bottom member AB is not equal to that of hypotenuse members OA and OB, which causes relative rotation between the members and affects the thermal deformation in the height direction after heating. In the same way, triangle OCD deforms the same as OAB. Triangles OAD and OBC consist of one ceramic component and do not cause rotation; therefore, they maintain an equilateral triangle. Thus, the members of the QP-structured metamaterial consist of different ceramics and rotate after heating, and the CTE value of the QP structure in the height direction can be programmatically adjusted from negative to positive.

According to the geometrical illustration (as shown in Fig. 3a), $\alpha_x = \alpha_2$, $\alpha_y = \alpha_1$, the characteristic lengths L_1 and L_2 are different, and the geometric relations can be expressed as

$$\begin{cases} L_1 = \dfrac{2H\cos\theta}{\sqrt{1-2\cos^2\theta}} \\ L_2 = \dfrac{H\sin\theta}{\sqrt{1-2\cos^2\theta}} \end{cases}, \tag{1}$$

where $L_1 = l_1$, and $L_2 = l_2\sin\theta$. With a temperature variation ΔT uniformly distributed on the QP-structured metamaterial, L_1 and L_2 deform. The differential of Eq. (1) leads to

$$\begin{cases} \dfrac{dL_1}{dT} = 2\left[\dfrac{\cos\theta}{\sqrt{1-2\cos^2\theta}}\cdot\dfrac{dH}{dT} - \dfrac{H\sin\theta}{(1-2\cos^2\theta)^{3/2}}\cdot\dfrac{d\theta}{dT}\right] \\ \dfrac{dL_2}{dT} = \dfrac{\sin\theta}{\sqrt{1-2\cos^2\theta}}\cdot\dfrac{dH}{dT} - \dfrac{H\cos\theta}{(1-2\cos^2\theta)^{3/2}}\cdot\dfrac{d\theta}{dT} \end{cases}, \tag{2}$$

According to the definition of the CTE [32], $\alpha = \frac{dl}{l\cdot dT}$; therefore, the CTE values of the QP-1-structured metamaterial in the x and y directions can be given as

$$\begin{cases} \alpha_1 = \alpha_H - \dfrac{\tan\theta}{1-2\cos^2\theta}\cdot\dfrac{d\theta}{dT} \\ \alpha_2 = \alpha_H - \dfrac{1}{\tan\theta(1-2\cos^2\theta)}\cdot\dfrac{d\theta}{dT} \end{cases}, \tag{3}$$

where $dL_i = \alpha_i L_i \cdot dT$ $(i = 1, 2)$. Then, the CTE value in the height direction can be calculated by

$$\alpha_z = \alpha_H = \frac{\alpha_2\sin^2\theta - \alpha_1\cos^2\theta}{1-2\cos^2\theta}. \tag{4}$$

In summary, the CTE values of the QP-1-structured metamaterial in the three directions are as follows:

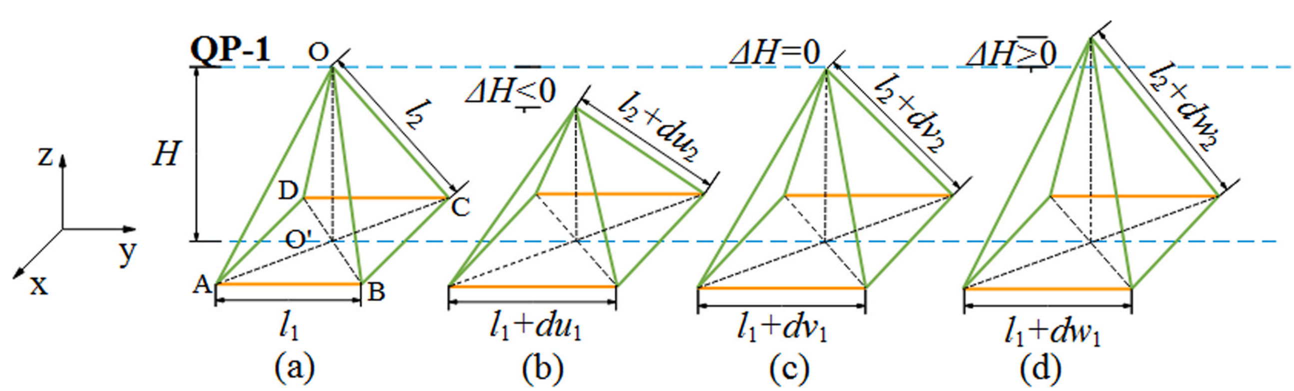

Fig. 3. Thermal expansion illustration of the multi-ceramic QP-1-structured metamaterial: (a) designed structure, (b) NTE behavior, (c) ZTE behavior, and (d) PTE behavior.

$$\begin{cases} \alpha_x = \alpha_2 \\ \alpha_y = \alpha_1 \\ \alpha_z = \dfrac{\alpha_2 \sin^2\theta - \alpha_1 \cos^2\theta}{1-2\cos^2\theta},\ \theta \in \left(45^\circ, 90^\circ\right). \end{cases} \tag{5}$$

2.1.2. QP-2-structured metamaterial

As discussed above, the QP-1-structured metamaterial was designed using different bottom members and the same hypotenuse members. Similarly, another structure, the QP-2 structure, can be designed by changing the two hypotenuse members, as shown in Fig. 4a. Square ABCD and triangle OAB consist of one ceramic; except for thermal expansion, the shape of square ABCD does not change after heating. The two hypotenuse members and the bottom member of isosceles triangle OCD consist of two ceramics. The members expand and rotate at vertex O after heating. At the same time, to cope with the thermal deformation of triangle OCD, the projection of vertex O to the bottom surface is no longer fixed at O′ but moves along the x-axis with the values of the material and geometric parameters of the QP-structured metamaterial.

Then, according to the geometric relationship, the characteristic lengths L_1 and L_2 can be expressed as

$$\begin{cases} L_1 = \dfrac{\sqrt{2}H\cos\theta}{\sqrt{1-2\cos^2\theta}} \\ L_2 = \dfrac{H}{\sqrt{1-2\cos^2\theta}} \end{cases}, \tag{6}$$

where $L_1 = l_1$, and $L_2 = l_2 \sin\theta$. The differential of Eq. (6) leads to

$$\begin{cases} \dfrac{dL_1}{dT} = 2\left[\dfrac{\cos\theta}{\sqrt{1-2\cos^2\theta}}\cdot\dfrac{dH}{dT} - \dfrac{H\sin\theta}{\left(1-2\cos^2\theta\right)^{3/2}}\cdot\dfrac{d\theta}{dT}\right] \\ \dfrac{dL_2}{dT} = \dfrac{\sin\theta}{\sqrt{1-2\cos^2\theta}}\cdot\dfrac{dH}{dT} - \dfrac{H\cos\theta}{\left(1-2\cos^2\theta\right)^{3/2}}\cdot\dfrac{d\theta}{dT} \end{cases}, \tag{7}$$

The CTE values of the QP-2-structured metamaterial in the x and y directions can be given as follows:

$$\begin{cases} \alpha_{L1} = \alpha_H - \dfrac{\tan\theta}{1-2\cos^2\theta}\cdot\dfrac{d\theta}{dT} \\ \alpha_{L2} = \alpha_H - \dfrac{1}{\tan\theta\left(1-2\cos^2\theta\right)}\cdot\dfrac{d\theta}{dT} \end{cases}, \tag{8}$$

Hence,

$$\alpha_z = \alpha_H = \frac{\alpha_{L2}\sin^2\theta - \alpha_{L1}\cos^2\theta}{1-2\cos^2\theta}, \tag{9}$$

where $\alpha_{L1} = \alpha_1$. α_{L2} is the CTE of isosceles triangle OBC in the height direction (as shown in Fig. 4e) and can be expressed as

$$\alpha_{L2} = \left\{\left[l_2^2\left(l_1^2+l_2^2\right)\alpha_1 + l_1^2\left(l_2^2+l_2^2\right)\alpha_2 + l_2^2\left(l_1^2+l_2^2\right)\alpha_2\right] - \left[2\cdot l_1^2 L_2^2\alpha_2 + l_2^4\alpha_1 + l_1^4\alpha_2 + l_2^4\alpha_2\right]\right\}\cdot\frac{1}{2\cdot l_1^2 L_2^2}, \tag{10}$$

Thus, substituting $OO'' = L_2 = l_2\sin\theta$ and $l_1 = 2l_2\cos\theta$ into Eq. (10) yields

$$\alpha_{L2} = \frac{\alpha_2\left(1-2\cos^2\theta\right)+\alpha_1}{2\sin^2\theta}, \tag{11}$$

Substituting Eq. (11) into (9), the CTE of the QP-2-structured metamaterial in the height direction can be expressed as

$$\alpha_z = \frac{\alpha_2\left(1-4\cos^2\theta\right)+\alpha_1}{2\left(1-2\cos^2\theta\right)}. \tag{12}$$

Consequently, the CTE values of the QP-2-structured metamaterial in the three directions can be given as

$$\begin{cases} \alpha_x = \alpha_y = \alpha_2 \\ \alpha_z = \dfrac{\alpha_1 + \alpha_2\left(1-4\cos^2\theta\right)}{2\left(1-2\cos^2\theta\right)}, \theta \in \left(45^\circ, 90^\circ\right). \end{cases} \tag{13}$$

2.1.3. QP-3-structured metamaterial

The above structures have different bottom members or hypotenuse members. Therefore, the QP-3-structured metamaterial was also designed using the same bottom members and hypotenuse members, as shown in Fig. 5a. Square ABCD consists of one ceramic; when it expands, its shape has no change after heating, and $\alpha_x = \alpha_y = \alpha_1$. Nevertheless, in multi-ceramic isosceles triangle OAB, the CTE values of bottom member AB and hypotenuse members OA and OB are not equal; therefore, these members rotate after heating. Then, the four bottom members expand in the horizontal directions and force the height of the QP-3-structured metamaterial to decrease after heating; likewise, the height increases with the expansion of the four hypotenuse members. Considering the symmetry of the QP structure, vertex O moves up, remains fixed, or moves down along OO', and the corresponding structure undergoes positive, zero, or negative expansion behavior after heating, respectively, as shown in Fig. 5b, c, and d.

For QP-3, l_1 and l_2 can be expressed as

$$\begin{cases} l_1 = \dfrac{\sqrt{2}H\cos\theta}{\sqrt{1-2\cos^2\theta}} \\ l_2 = \dfrac{H}{\sqrt{1-2\cos^2\theta}} \end{cases}, \tag{14}$$

The differential of Eq. (14) leads to

$$\begin{cases} \dfrac{dl_1}{dT} = \sqrt{2}\left[\dfrac{\cos\theta}{\sqrt{1-2\cos^2\theta}}\cdot\dfrac{dH}{dT} - \dfrac{H\sin\theta}{\left(1-2\cos^2\theta\right)^{3/2}}\cdot\dfrac{d\theta}{dT}\right] \\ \dfrac{dl_2}{dT} = \dfrac{1}{\sqrt{1-2\cos^2\theta}}\cdot\dfrac{dH}{dT} - \dfrac{2H\sin\theta\cos\theta}{\left(1-2\cos^2\theta\right)^{3/2}}\cdot\dfrac{d\theta}{dT} \end{cases}, \tag{15}$$

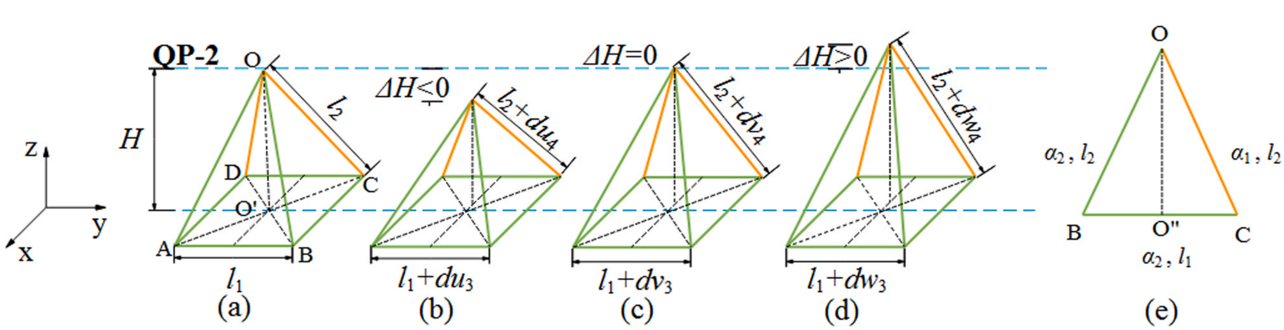

Fig. 4. Thermal expansion illustration of the multi-ceramic QP-2-structured metamaterial: (a) designed structure, (b) NTE behavior, (c) ZTE behavior, (d) PTE behavior, and (e) schematic of isosceles triangle OBC.

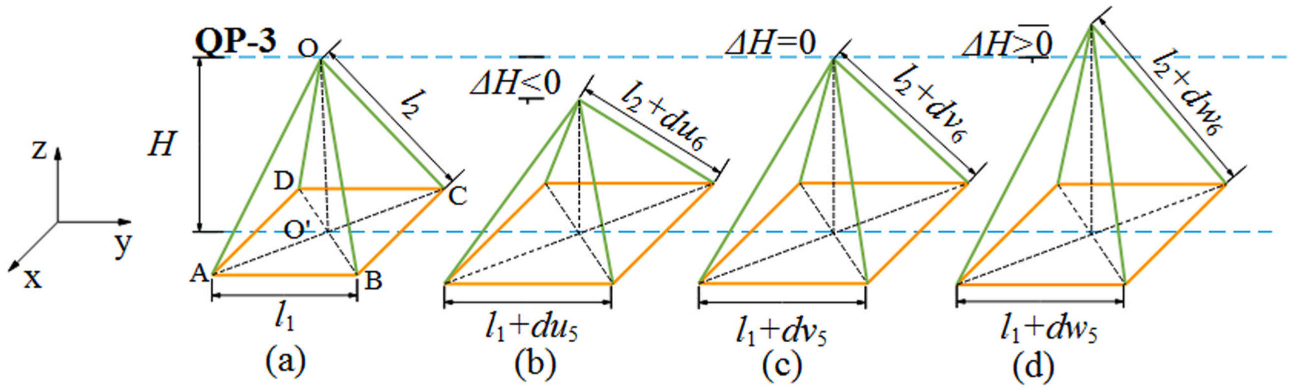

Fig. 5. Thermal expansion illustration of the multi-ceramic QP-3-structured metamaterial: (a) designed structure, (b) NTE behavior, (c) ZTE behavior, and (d) PTE behavior.

Therefore,

$$\begin{cases} \alpha_1 = \alpha_H - \dfrac{\tan\theta}{1-2\cos^2\theta}\cdot\dfrac{d\theta}{dT} \\ \alpha_2 = \alpha_H - \dfrac{1}{\tan\theta(1-2\cos^2\theta)}\cdot\dfrac{d\theta}{dT} \end{cases}, \tag{16}$$

The CTE value of the QP-3-structured metamaterial in the height direction can be represented as

$$\alpha_z = \alpha_H = \frac{\alpha_2 - 2\alpha_1\cos^2\theta}{1-2\cos^2\theta}. \tag{17}$$

Thus, the CTE values of the QP-3-structured metamaterial in the three directions can be given as

$$\begin{cases} \alpha_x = \alpha_y = \alpha_1 \\ \alpha_z = \dfrac{\alpha_2 - 2\alpha_1\cos^2\theta}{1-2\cos^2\theta}, \theta \in (45^\circ, 90^\circ). \end{cases} \tag{18}$$

2.2. Thermal expansion analysis

2.2.1. QP-1-structured metamaterial

In this study, Al_2O_3 and ZrO_2 ceramics were used as the constituent materials for 3D-architected ceramic metamaterials. Al_2O_3 and ZrO_2 ceramics are promising materials for stereolithography additive manufacturing [32–38]. The CTE values of the stereolithography additively manufactured Al_2O_3 and ZrO_2 ceramics are 8.20×10^{-6}/°C and 12.78×10^{-6}/°C, respectively. For the QP-1 structure (as given in Fig. 2a), $\alpha_1 > \alpha_2$ when the orange members and green members are ZrO_2 and Al_2O_3, respectively. Using Eq. (5), the CTE value of the QP-1-structured metamaterial in the height direction (α_z) can be obtained, as shown in Fig. 6a. The CTE value in the height direction is strongly dependent on the angle θ. It can be observed that α_z increases with increasing θ. The CTE can be tailored from a negative value to a positive value. Fig. 3b shows the QP-1-structured metamaterial with NTE behavior. Because both the ZrO_2 and Al_2O_3 ceramics have positive CTE values, both du_1 and du_2 are above 0. ΔH is below 0 when the value of θ is above 51.27°, indicating that the height of the QP-1-structured metamaterial decreases after heating. Thus, the QP-1-structured metamaterial exhibits NTE behavior. Moreover, the QP-1-structured metamaterial can exhibit ZTE behavior, as shown in Fig. 3c. When the value of θ is set at 51.27°, both dv_1 and dv_2 are above zero, and ΔH remains zero; this means that the height of the QP-1 structure does not change after heating, revealing that the QP-1 structure exhibits ZTE behavior. Apart from the NTE and ZTE behaviors, PTE behavior of the QP-1-structured metamaterial can also be achieved when the value of θ is above 51.27°, as shown in Fig. 3d. Both dw_1 and dw_2 are above zero, and ΔH is also above zero; therefore, the height change of the QP-1-structured metamaterial is positive. It should be noted that the tailoring range for PTE is relatively narrow when the orange members and green members are ZrO_2 and Al_2O_3, respectively.

Fortunately, a large tailoring range for PTE can be achieved by using Al_2O_3 as the two bottom members and ZrO_2 both as the two bottom members and the four hypotenuse members. Both dw_1 and dw_2 are above zero, and because α_1 is lower than α_2, ΔH is also above zero, as shown in Fig. 3d. Thus, the CTE value in the height direction (α_z), which is substantially larger than the CTE value of the ZrO_2 ceramic, always maintains a positive value and gradually increases with increasing θ, as shown in Fig. 6b.

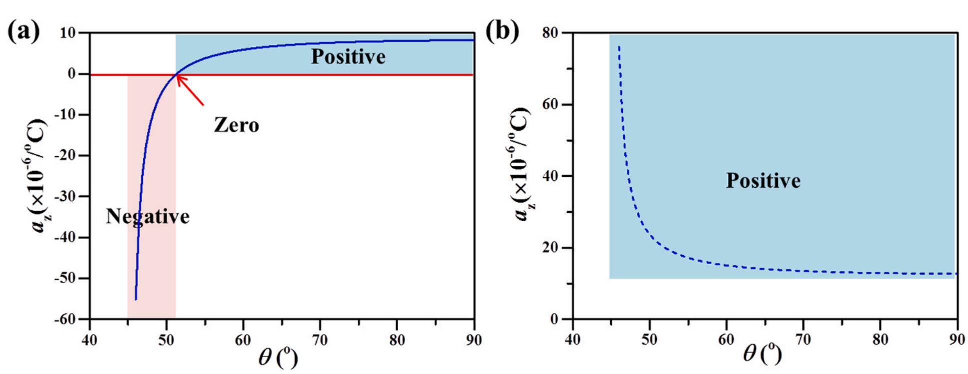

Fig. 6. Thermal expansion values in the height direction (α_z) *vs.* the angle (θ) of the QP-1-structured metamaterial: (a) $\alpha_1 > \alpha_2$: the orange and green lines are ZrO_2 and Al_2O_3, respectively; (b) $\alpha_1 < \alpha_2$: the orange and green lines are Al_2O_3 and ZrO_2, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2.2.2. QP-2-structured metamaterial

For the QP-2-structured metamaterial, the first condition is when the four bottom members and two hypotenuse members are Al_2O_3, and the other two hypotenuse members (both OC and OD) are ZrO_2. Under these conditions, the CTE values of both the four bottom members and the two hypotenuse members (green line) are higher than those of the other two hypotenuse members (orange line). Therefore, according to Eq. (12), the CTE value of the QP-2-structured metamaterial in the height direction remains positive, as shown in Fig. 7a. It can be observed that as θ increases, α_z decreases. The QP-2-structured metamaterial shows PTE behavior after heating, as shown in Fig. 4d.

For the second condition, the four bottom members and the two hypotenuse members consist of ZrO_2, and the other two hypotenuse members are Al_2O_3. Using Eq. (12), the CTE value in the height direction (α_z) *vs.* the angle (θ) of the QP-2-structured metamaterial can be obtained, as shown in Fig. 7b. Under these conditions, both NTE and PTE behaviors can be obtained. As the angle (θ) increases, the QP-2-structured metamaterial exhibits NTE behavior first (as shown in Fig. 4b) and then shows ZTE behavior at an angle of 50.15° (as shown in Fig. 4c). When the angle θ is above 50.15°, the QP-2-structured metamaterial displays PTE behavior and is slightly lower than 10.49×10^{-6}/°C (as shown in Fig. 4d).

2.2.3. QP-3-structured metamaterial

The QP-3-structured metamaterial is different from the QP-1- and QP-2-structured metamaterials. The bottom members and hypotenuse members consist of two different ceramics. When the bottom members are ZrO_2 and the hypotenuse members are Al_2O_3, the CTE value in the height direction (α_z) *vs.* the angle (θ) of the QP-3 structure can be obtained using Eq. (17), as shown in Fig. 8a. With an increase in the angle (θ), the CTE value in the height direction (α_z) increases sharply, and the QP-3 structure shows NTE behavior (as shown in Fig. 5b). When the angle (θ) is 55.5°, the QP-3 structure displays ZTE behavior (as shown in Fig. 5c). Moreover, the CTE value in the height direction increases slowly, and the QP-3 structure exhibits PTE behavior when the angle (θ) is higher than 55.5° however, the highest CTE value is still lower than that of the pristine Al_2O_3 ceramic (as shown in Fig. 8a).

In contrast, the bottom members and the hypotenuse members can also consist of Al_2O_3 and ZrO_2, respectively. Under this condition, a wide range of PTE is available. According to Eq. (17), the CTE value in the height direction (α_z) always remains positive and decreases gradually as the angle (θ) increases and is also larger than the CTE value of the pristine ZrO_2 ceramic, as shown in Fig. 8b.

2.3. Design results and analysis

In summary, the thermal expansion behavior of the 3D-architected multi-ceramic metamaterials strongly depends on both the angle (θ) and the CTE values of the two ceramics. Indeed, the ranges of tailorable CTE values of the three 3D-architected multi-ceramic metamaterials in the height direction are different. In theory, when the angle (θ) approaches 45°, the CTE approaches ∞. For comparison, an angle (θ) greater than or equal to 46° was chosen. The CTE value in the height direction (α_z) *vs.* the angle (θ) of the three 3D-architected multi-ceramic metamaterials is shown in Fig. 9a. When $\alpha_1 < \alpha_2$ (the orange line is Al_2O_3, and the green line is ZrO_2), the CTE values of the QP-1- and QP-3-structured metamaterials are both greater than 0; that is, only PTE can be achieved. However, the QP-2-structured metamaterial can realize thermal expansion from negative to positive. In contrast, when $\alpha_1 > \alpha_2$ (the orange member is ZrO_2, and the green member is Al_2O_3), the QP-1- and QP-3-structured metamaterials can adjust the thermal expansion from negative to positive, and the QP-2 metamaterial can only achieve PTE. The ranges of tailorable CTE values of the three multi-ceramic structures in the height direction are summarized in Fig. 9b. It can be seen that the ranges of tailorable CTE values of the QP-1-, QP-2-, and QP-3-structured metamaterials in the height direction are (−55.12 - +76.11) $\times 10^{-6}$/°C, (−52.84 - +73.81) $\times 10^{-6}$/°C and (−118.15 - +139.44) $\times 10^{-6}$/°C, respectively. It should be noted that the QP-1- and QP-3-structured metamaterials cannot achieve CTEs between 8.20×10^{-6}/°C and 12.78×10^{-6}/°C.

When the geometric parameter angle (θ) is constant, the tailorable CTE range of the QP-3-structured metamaterial is the largest, followed by that of the QP-1-structured metamaterial; that of the QP-2-structured metamaterial is the smallest. This is because the CTE values of the two hypotenuse members of the QP-2-structured metamaterial are different from those of the other two, which causes the hypotenuse members to deform both in the height direction and in the in-plane direction after heating, which then causes the QP-2-structured metamaterial to deform in the height direction. Thus, the absolute CTE value of the QP-2-structured metamaterial in the height direction is smaller. Moreover, compared to the QP-1-structured metamaterial, the QP-3-structured metamaterial has four members with high CTE values at the bottom, which causes a larger thermal mismatch after heating. The rotation between the pieces is greater, which in turn produces greater thermal deformation.

In summary, the tailorable CTE range of the QP-3-structured metamaterial is the largest. Therefore, to verify that the design is feasible, the QP-3 structure was chosen to fabricate the 3D-architected multi-ceramic metamaterials in this study. The tailorable CTE range of the QP-3-structured metamaterial range from 8.20×10^{-6}/°C to 12.78×10^{-6}/°C, which can be designed by the modified Kerner model [39].

$$\alpha^c = V_f \alpha_f^c + V_m \alpha_m^c \tag{19}$$

where c is a constant power. The value of c is empirically determined, and $c = 1$ was chosen in this study. In other words, the CTE range from

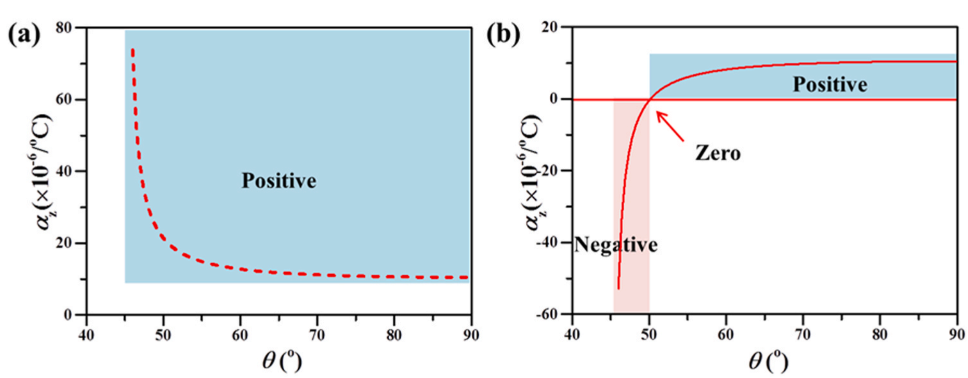

Fig. 7. Thermal expansion value in the height direction (α_z) *vs.* the angle (θ) of the QP-2-structured metamaterial: (a) $\alpha_1 > \alpha_2$: the orange and green lines are Al_2O_3 and ZrO_2, respectively; (b) $\alpha_1 < \alpha_2$: the orange and green lines are ZrO_2 and Al_2O_3, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

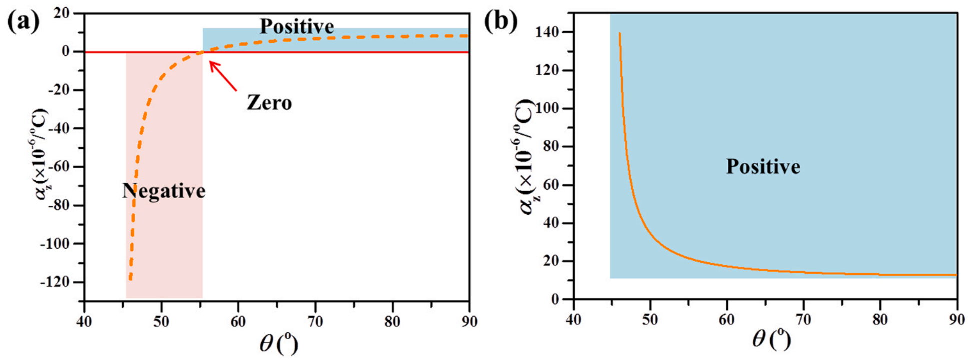

Fig. 8. Thermal expansion value in the height direction (α_z) *vs.* the angle (θ) of the QP-3-structured metamaterial: (a) $\alpha_1 > \alpha_2$: the orange and green lines are ZrO_2 and Al_2O_3, respectively; (b) $\alpha_1 < \alpha_2$: the orange and green lines are Al_2O_3 and ZrO_2, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

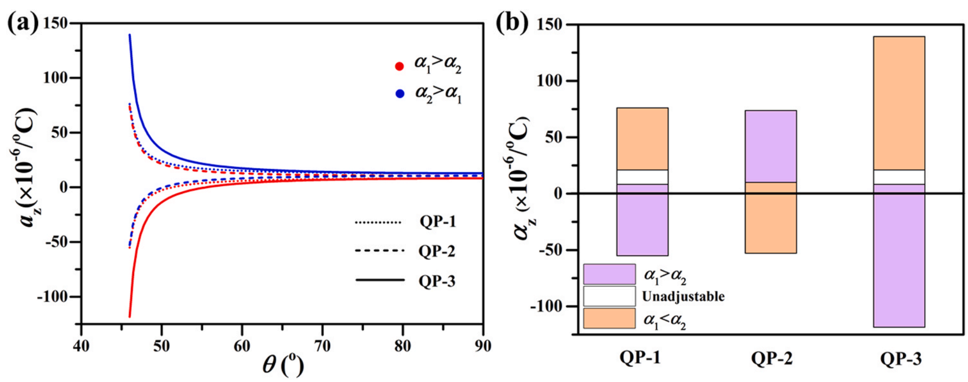

Fig. 9. (a) Thermal expansion value in the height direction (α_z) *vs.* the angle (θ) of the three structured metamaterials; (b) tailoring ranges of the CTE for the three 3D-architected multi-ceramic metamaterials.

8.20×10^{-6}/°C to 12.78×10^{-6}/°C can be achieved by changing the volume ratio of ZrO_2 and Al_2O_3. These structures consist of homogeneous Al_2O_3/ZrO_2 composition. It should be noted that the angle (θ) is arbitrary owing to its homogeneous composition. The CTEs in the different directions are equal.

To verify the reliability of this design method, multi-ceramic QP-3-structured parts with NTE, ZTE, and PTE behaviors were designed, stereolithography additively manufactured, and characterized. The CTE values in the height direction (α_z) of the multi-ceramic QP-3-structured parts with NTE, ZTE, and PTE behaviors were chosen to be -10×10^{-6}/°C, 0×10^{-6}/°C and $+10 \times 10^{-6}$/°C, respectively. Accordingly, the respective angles (θ) were calculated to be 50.8°, 55.5° and 35.5° according to Eqs. (18) and (19). It should be noted that the volume ratio of ZrO_2 to Al_2O_3 in the multi-ceramic QP-3-structured part with a CTE of $+10 \times 10^{-6}$/°C was 39.3:60.7, according to Eq. (19). The detailed structural design data are listed in Table 1. 3D models of the three designed multi-ceramic QP-3-structured metamaterials with NTE, ZTE, and PTE behaviors were drawn using Solidworks software, as shown in Fig. 10a$_1$, b$_1$, and c$_1$, respectively.

Table 1
Design of the QP-3-structured parts with NTE, ZTE, and PTE behaviors.

	Programmable thermal expansion		
	NTE	ZTE	PTE
Bottom	ZrO_2	ZrO_2	Al_2O_3/ZrO_2
Hypotenuse	Al_2O_3	Al_2O_3	Al_2O_3/ZrO_2
θ (°)	50.8	55.5	35.5
Designed α_z (×10^{-6}/°C)	-10.00	0	10.00
Measured α_z (×10^{-6}/°C)	-8.87	0.81	9.45

3. Stereolithography additive manufacturing

3.1. Raw materials

Al_2O_3 powders: Coarse Al_2O_3 (c-Al_2O_3, d_{50} = 10.34 μm) and fine Al_2O_3 (f-Al_2O_3, d_{50} = 1.05 μm), supplied by Zhongzhou Alloy Material Co., Ltd., China, were used as the raw powders with a volume ratio of 85:15 to prepare the Al_2O_3 slurry.

ZrO_2 powders: 3Y-TZP tetragonal ZrO_2 (t-ZrO_2, d_{50} = 0.2 μm, Jiangxi Size Material Co., Ltd., China) and monoclinic ZrO_2 (m-ZrO_2, d_{50} = 0.2 μm, Zhongzhou Alloy Material Co., Ltd., China) were used as raw powders with a volume ratio of 85:15 to fabricate the ZrO_2 slurry.

TiO_2 (anatase, 60 nm, Aladdin, China) and MgO (50 nm, Aladdin, China) were employed as sintering additives to improve the densification and mechanical properties.

TMPTA (1.1 g/cm^3, Changshu Hengrong Trading Co., Ltd., China) and HDDA (1.01 g/cm^3, Aladdin, China) with a volume ratio of 1:4 were used as the diluent and photosensitive resin, respectively, to prepare the photosensitive slurries. TPO (Guangzhou Lihou Trading Co., Ltd., China) was added as a radical photoinitiator. The polymeric compound KOS110 (Guangzhou Kangoushuang Trade Co., Ltd., China) was used as a dispersant to improve the fluidity and printability of the slurries.

3.2. Slurry preparation

The slurry used for stereolithography additive manufacturing was first prepared. Similar to the 2D-architected ceramic metamaterials reported in our previous work [31], 3D-architected multi-ceramic QP-3-structured metamaterial with NTE and ZTE behaviors were additively manufactured from a 55 vol% Al_2O_3 slurry and 50 vol% ZrO_2

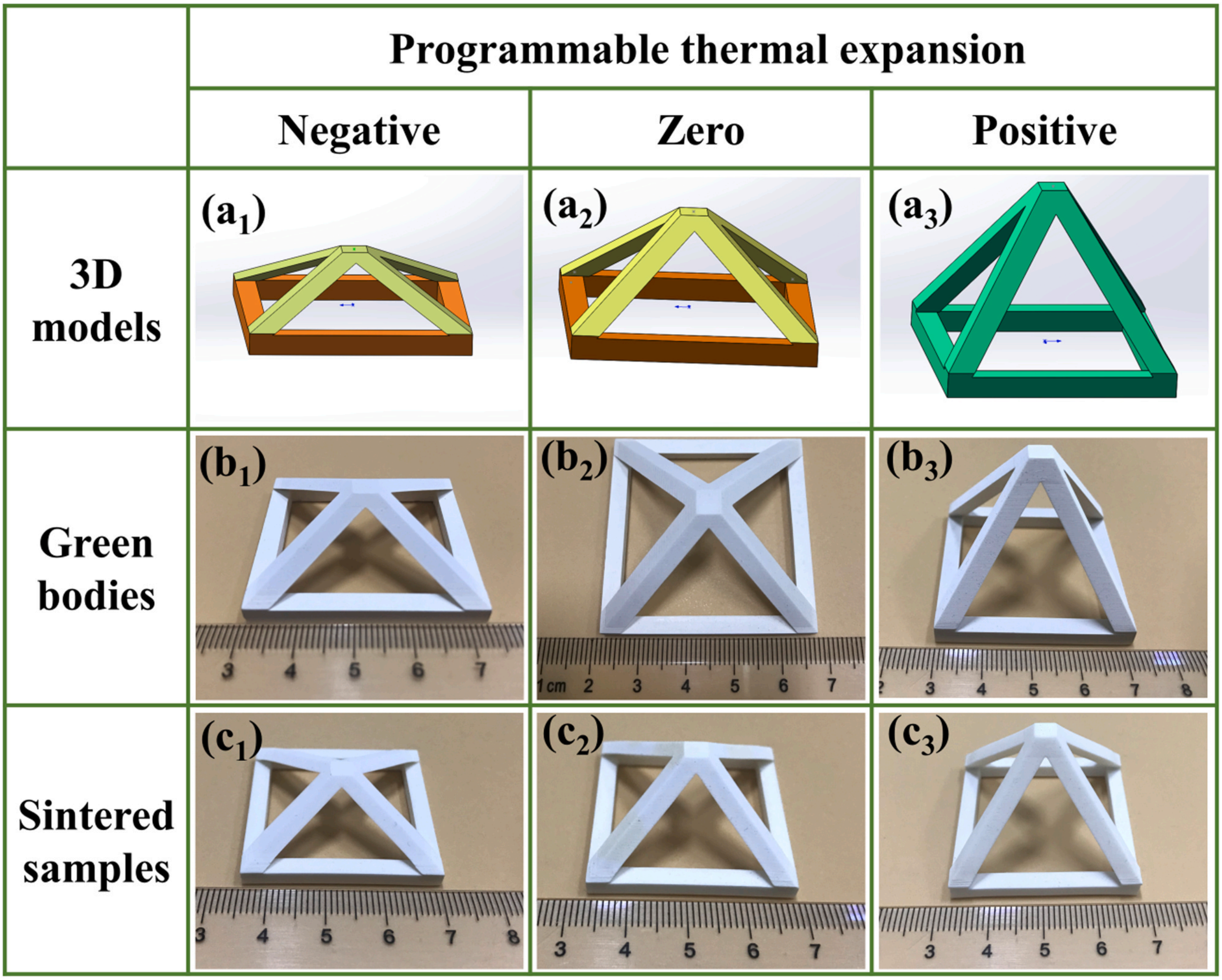

Fig. 10. (a) 3D model, (b) as-prepared green bodies, and (c) sintered samples of the designed multi-ceramic QP-3 structures.

slurry, respectively. The two slurry compositions are given in our previous work [40,41]. In addition, a mixed slurry with a volume ratio of 39.3:60.7 was prepared using ZrO_2 and Al_2O_3 powders, which was then used to fabricate the 3D-architected multi-ceramic QP-3-structured metamaterial green body with PTE behavior. The mixtures were ball-milled for 24 h in a planetary mill (Qmonoclinic3SP2, Nanjing University Instrument Plant, China) at 400 rpm. Subsequently, three FGC slurries, 75Zr–25Al, 50Zr–50Al, and 25Zr–75Al, were prepared from the 55 vol% Al_2O_3 and 50 vol% ZrO_2 slurries, according to our previous report [31].

3.3. Additive manufacturing

Stereolithography additive manufacturing was subsequently conducted. Stereolithography additive manufacturing equipment with an ultraviolet light wavelength of 405 nm (AutoCera, Beijing 10Dim Tech. Co., Ltd., China) was used to manufacture 3D-architected multi-ceramic QP-3-structured green bodies with NTE and ZTE behaviors. First, the 3D model drawn by Solidworks software was sliced using 10dim software with a slicing thickness of 50 μm. The exposure times of the primary layer and other layers were set to 50 s and 4 s, respectively. Photosensitive ZrO_2 (50 vol%) slurry was poured into the vat. Then, the four bottom members were manufactured, and the ZrO_2 slurry was removed from the vat. Subsequently, the FGC layers, 75Zr–25Al, 50Zr–50Al, and 25Zr–75Al, were manufactured sequentially, as reported in a previous study [31]. Four hypotenuse members were then manufactured. Finally, 3D-architected multi-ceramic QP-3-structured green bodies with NTE and ZTE behaviors were obtained and then ultrasonically washed with ethanol to remove the uncured slurry, as shown in Fig. 10b$_1$ and b$_2$. In addition, a 3D-architected multi-ceramic QP-3-structured green body with PTE behavior was printed from an Al_2O_3–ZrO_2 mixed slurry (the volume ratio of ZrO_2 to Al_2O_3 was 39.3:60.7) without an FGC layer, as shown in Fig. 10b$_3$. These green bodies showed good surface quality, no defects, and a well-defined morphology, illustrating that stereolithography additive manufacturing is suitable for the fabrication of 3D-architected multi-ceramic metamaterials.

3.4. Debinding and sintering

To remove the organic matter, including photosensitive resin, photoinitiator, and dispersant, the as-prepared 3D-architected multi-ceramic green bodies were debinded in a furnace (FMJ-07/11, HeFei Facerom Thermal Co., Ltd., China) at 600 °C for 2 h and pre-sintered at 1000 °C for 2 h in air, as reported in our previous work [41]. The pre-sintered samples were subsequently transported and then sintered in a high-temperature furnace (FMJ-05/17, HeFei Facerom Thermal Co., Ltd., China) at 1600 °C for 2 h in air. Detailed debinding and sintering processes were also reported in our previous work [41]. The as-prepared green bodies showed shrinkage owing to the use of a photosensitive resin. Therefore, a slow heating rate and high homogeneity were used in this study. In addition, FGC layers were employed to reduce and overcome the thermal mismatch between the two constituent ceramics.

Therefore, the shrinkage of the as-prepared green body did not cause a crack in the multi-ceramic QP-3-structured metamaterial sample. Finally, defect-free sintered samples were obtained, as shown in Fig. 10c_1, c_2, and c_3.

4. Thermal expansion behavior characterization

4.1. Testing method and system

As shown in Fig. 11, the thermal expansion of the multi-ceramic QP-3-structured metamaterial was measured using a custom-built testing system, which was developed based on non-contact digital image correlation (DIC) technology. DIC technology was originally conceived and developed by researcher in Japan and the United States in the 1980s. It can be measured the deformations incurred by a nominally planar object, which is subjected to loads leading to mainly "in-plane" motion [42–44]. The principle is to obtain the displacement vector of the pixel by tracking (or matching) the position of the same pixel in the two speckle images before and after the object surface is deformed, thereby obtaining the full-field displacement of the specimen surface [45,46]. In this study, white light and curtains were used to eliminate environmental noise and improve image quality. Before heating, the sintered sample surface was painted with white speckles using high-temperature paint, and then two diagonal vertices of the bottom quadrilateral and vertices of the quadrangular pyramid were coated with black speckles, as shown in Fig. 11. The painted sample was then transformed and laid flat in a furnace. A high-resolution camera (Ximea MD091MU-SY 14 bit black/white 3380 × 2704 progressive scan CCD) was used to obtain reference images of the painted sample at room temperature. Subsequently, the painted sample was uniformly heated to 700 °C. When the temperature reached 100, 200, 300, 400, 500, 600, and 700 °C, images of the painted sample were recorded using the CCD camera. Finally, using DIC technology with the help of the weighted zero normalized difference of squares (ZNSSD) correlation criterion [47], the displacement cloud maps of the QP-3-structure metamaterial can be obtained and the thermally induced displacements of the samples can be then calculated with the temperature changing from the displacement cloud maps.

4.2. Testing results and analysis

As mentioned above, the thermal expansion of the multi-ceramic QP-3-structured metamaterials in the height direction was measured using a custom-built testing system, as shown in Fig. 11. Images of the multi-ceramic QP-3-structured parts at room temperature and various elevated temperatures were captured using a CCD camera and used for DIC calculation, as shown in Fig. 12. Fig. 12a_0-a_7 show images of the multi-ceramic QP-3-structured sample with NTE behavior at room temperature and 100, 200, 300, 400, 500, 600, 700 °C, respectively. Fig. 12b_0-b_7 present images of the multi-ceramic QP-3-structured sample with ZTE behavior at room temperature and 100, 200, 300, 400, 500, 600, 700 °C, respectively. Fig. 12c_0-c_7 show images of the multi-ceramic QP-3-structured sample with PTE behavior at room temperature and 100, 200, 300, 400, 500, 600, 700 °C, respectively. These images were used to calculate the CTE values of the multi-ceramic QP-3 structures. The position coordinate information of the black speckles was recorded by a CCD camera and analyzed by DIC. Then, the displacements of the multi-ceramic QP-3-structured samples in the height direction were obtained according to the position coordinate information of the black speckles at room temperature and various test temperatures. Subsequently, the thermal strain, ε, was calculated as the temperature changed. Finally, the equivalent thermal expansion values of the multi-ceramic QP-3-structured samples in the height direction were calculated using Eq. (20).

$$\alpha_z = \frac{\varepsilon}{\Delta T}, \tag{20}$$

where ε is the thermal strain, and ΔT is the temperature change of the multi-ceramic QP-3-structured samples.

In Fig. 13, both the DIC calculated values and the linear fit of the thermal strain with the change in temperature are plotted. From Fig. 13a, it can be seen that the DIC calculated thermal strain of the as-designed multi-ceramic QP-3 structure with a θ of 50.8° was less than zero and decreased as the temperature increased, illustrating that the structure exhibited NTE behavior. Surprisingly, as shown in Fig. 13b, the DIC calculated thermal strain of the as-designed multi-ceramic QP-3 structure with a θ of 55.5° was close to zero and only changed slightly with increasing temperature, demonstrating that the structure exhibited zero or near-zero thermal expansion behavior. However, the DIC calculated thermal strain of the as-designed multi-ceramic QP-3-structured metamaterial with a θ of 35.5° was larger than zero and increased with increasing temperature, indicating that the structure exhibited PTE behavior, as shown in Fig. 13c.

Eq. (20) was used to calculate the thermal expansion value (α_z) in the height direction of the multi-ceramic QP-3-structured metamaterials, where α_z can be obtained by calculating the slope of the linear fit of the DIC calculated thermal strain in Fig. 13. The calculated thermal expansion values (α_z) are summarized in Table 1. The calculated α_z values of the multi-ceramic structures with NTE, near-ZTE, and PTE behaviors were -8.87×10^{-6}/°C, 0.81×10^{-6}/°C, and 9.45×10^{-6}/°C, respectively. Both the experimentally measured values and designed values for the multi-ceramic QP-3-structured metamaterials are plotted

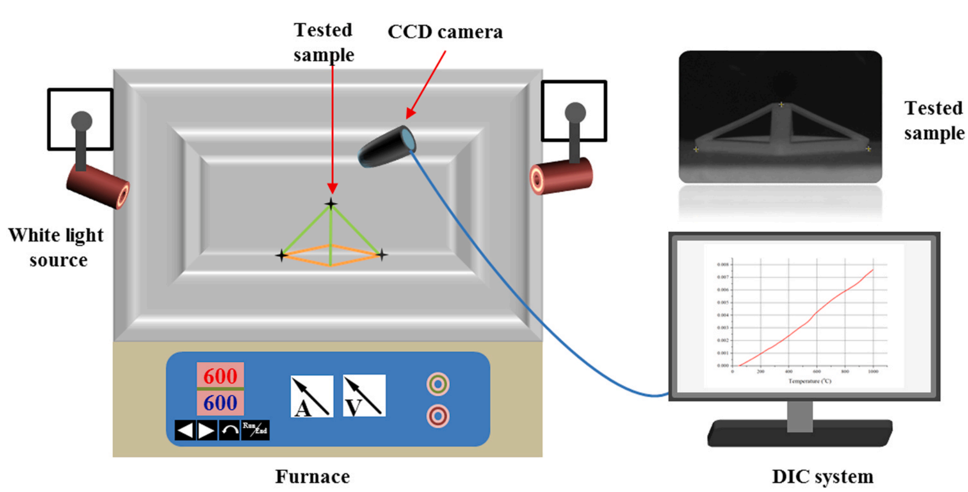

Fig. 11. Custom-built thermal expansion testing system for the multi-ceramic structured parts.

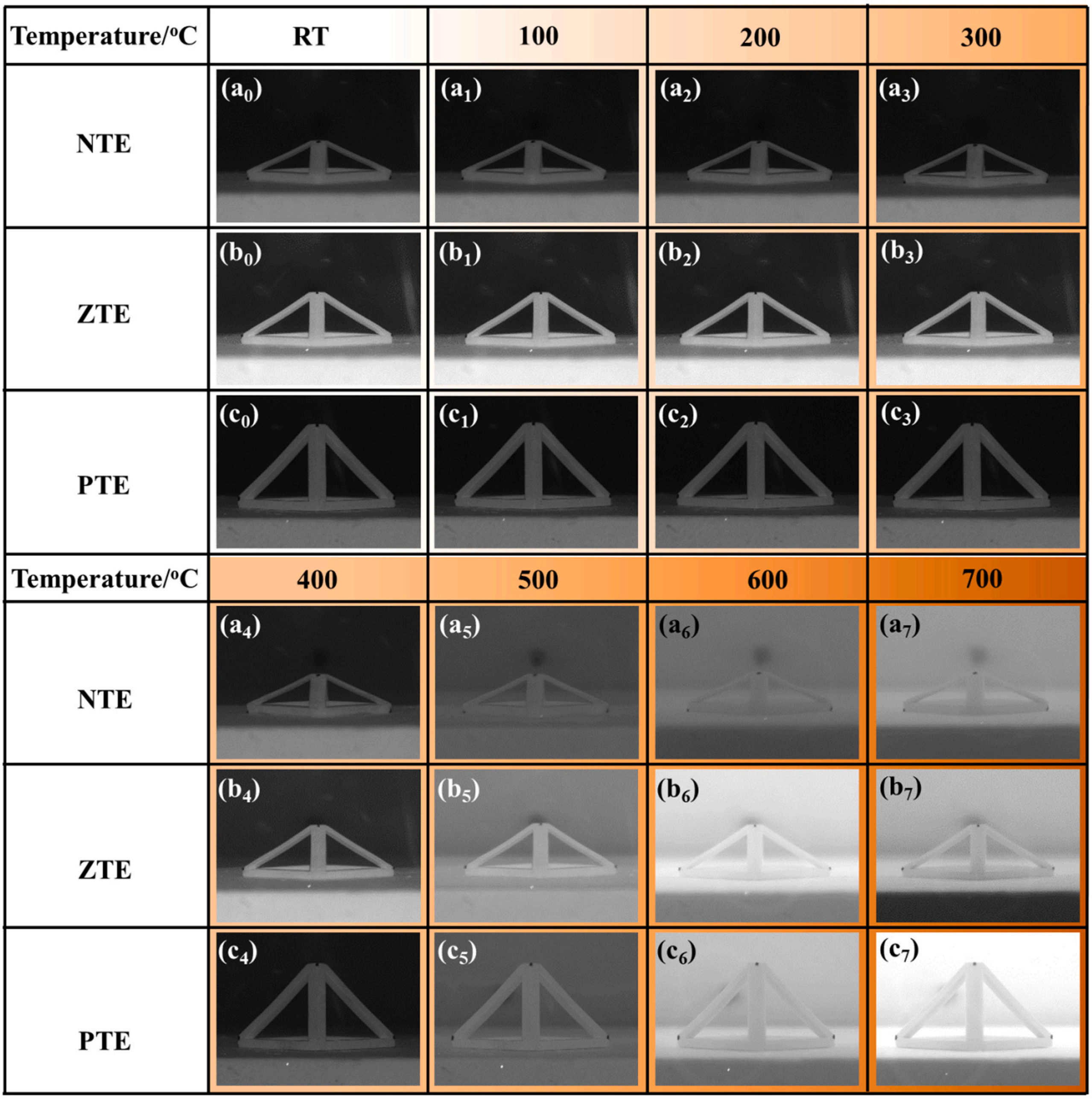

Fig. 12. Images of the multi-ceramic QP-3-structured metamaterials at room temperature and various test temperatures.

in Fig. 14, which clearly shows that the experimentally measured values of the multi-ceramic QP-3-structured parts were in good agreement with the designed values. Small errors of $1.13 \times 10^{-6}/°C$, $0.81 \times 10^{-6}/°C$, and $0.55 \times 10^{-6}/°C$ were found between the experimentally measured and designed values for the multi-ceramic QP-3-structured metamaterials with NTE, near-ZTE, and PTE behaviors, respectively, which were all below $1.5 \times 10^{-6}/°C$. The errors were mainly caused by two aspects: experimental error from the testing process and DIC calculation and error in the designed values. Generally, the CTEs of Al_2O_3 and ZrO_2 vary non-linearly with temperature [48–51]. To simplify the design, the CTEs used in this study were linearized. However, only small errors were found, and the experimental measurements were in good agreement with the theoretical design. These results also confirm the effectiveness of lightweight multi-ceramic structure design, the feasibility of the manufacturing process, and the wide range of programmable thermal expansion behavior.

5. Conclusions

In this investigation, 3D-architected multi-ceramic metamaterials with programmable thermal expansion, which could be used in high-temperature or large-temperature-range environments, were designed, fabricated, and characterized. The main results are summarized as follows:

(1) The theoretical basis for designing 3D-architected multi-ceramic metamaterials with programmable thermal expansion was presented. Three typical multi-ceramic quadrangular pyramid (QP)-structured metamaterials were designed and discussed.
(2) The thermal expansion behavior of the 3D-architected multi-ceramic metamaterials was analyzed. The ranges of tailorable CTE values of the QP-1-, QP-2-, and QP-3-structured metamaterials in the height direction were (−55.12 - +76.11) ×

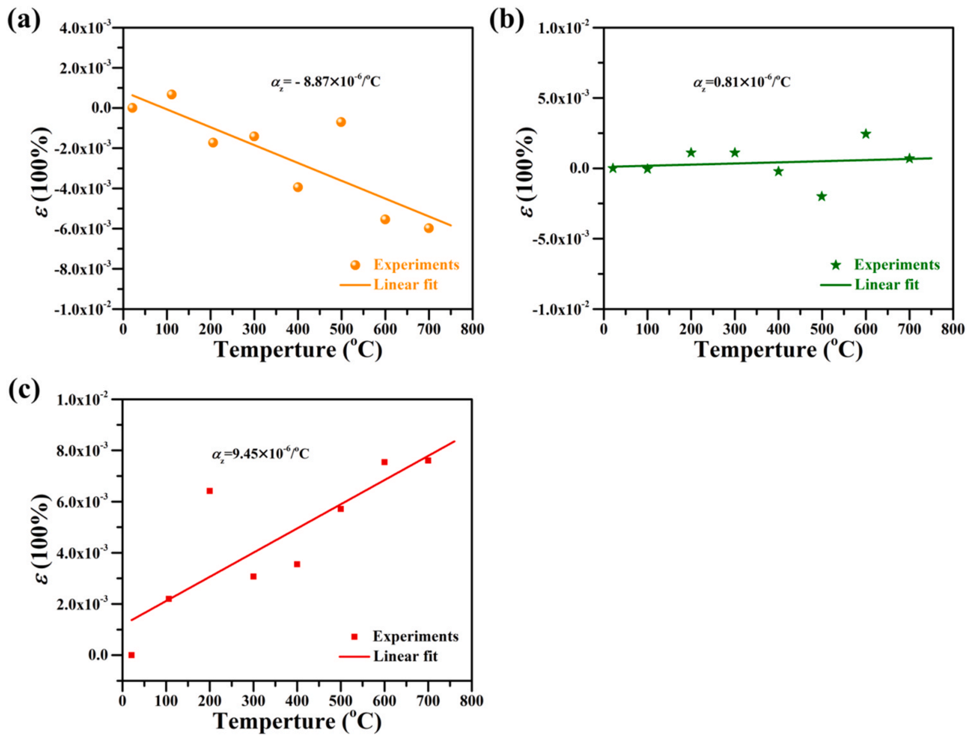

Fig. 13. Thermal strain of the multi-ceramic QP-3-structured metamaterials *vs.* temperature calculated by DIC: (a) designed θ of 50.8°, (b) designed θ of 55.5°, and (c) designed θ of 35.5°.

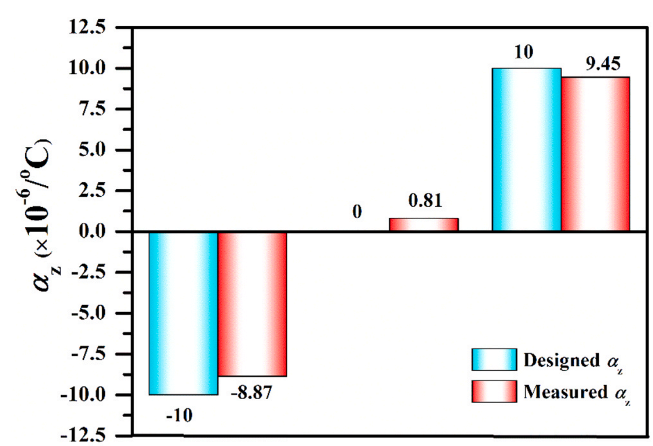

Fig. 14. Comparison between the experimentally measured and designed thermal expansion values of lightweight multi-ceramic structured parts.

10^{-6}/°C, (−52.84 - +73.81) × 10^{-6}/°C and (−118.15 - +139.44) × 10^{-6}/°C, respectively.

(3) The multi-ceramic QP-3-structured metamaterial was chosen and designed based on ZrO_2 and Al_2O_3 ceramics. These multi-ceramic QP-3-structured parts, which exhibited negative, zero, and positive thermal expansion behaviors with well-defined morphologies, were manufactured by stereolithography additive manufacturing, debinding, and sintering.

(4) The thermal expansion behaviors of the multi-ceramic metamaterial parts were analyzed using DIC. The calculated α_z values of the multi-ceramic metamaterials with negative, near-zero, and positive thermal expansion behaviors were − 8.87 × 10^{-6}/°C, 0.81 × 10^{-6}/°C, and 9.45 × 10^{-6}/°C, respectively. The experimental measurements were in good agreement with the theoretical design.

In summary, this study confirmed the effectiveness of 3D-architected multi-ceramic metamaterials with programmable thermal expansion design, the feasibility of the manufacturing process, and the wide range of programmable thermal expansion possible with this lightweight material.

CRediT authorship contribution statement

Keqiang Zhang: Conceptualization, Methodology, Investigation, Data curation, Formal analysis, Validation, Writing – original draft. **Kaiyu Wang**: Investigation, Validation, Data curation, Formal analysis, Writing – review & editing. **Jiaxin Chen**: Investigation, Validation, Data curation, Formal analysis, Writing – review & editing. **Kai Wei**: Investigation, Data curation, Methodology, Writing – review & editing, Project administration. **Bo Liang**: Investigation, Validation, Writing – review & editing, Project administration. **Rujie He**: Conceptualization, Methodology, Validation, Writing – review & editing, Funding acquisition, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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